

A Report on

**IOT BASED SMART SYSTEM FOR FALL
DETECTION OF ELDERLY PEOPLE**

For

Mini Project 2B (Autonomy R-2024 Scheme) of Third Year, (TY SEM-VI)

in

Electronics and Telecommunication Engineering

by

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ABSTRACT

The increasing number of fall incidents among elderly individuals has become a major concern in modern healthcare, especially for those living independently without constant supervision. Falls can lead to severe injuries, delayed medical assistance, and life-threatening complications if not detected in time. Therefore, there is a growing need for a reliable, real-time fall detection system that can ensure immediate response and enhance the safety of elderly individuals.

This project presents an IoT-based wearable smart watch designed for real-time fall detection and health monitoring. The system is built using an ESP32-S3 microcontroller integrated with an IMU sensor (accelerometer and gyroscope) to continuously monitor the user's motion and detect sudden changes in acceleration and orientation that may indicate a fall. To improve detection accuracy and reliability, a MAX30102 sensor is incorporated to measure physiological parameters such as heart rate and blood oxygen level (SpO₂), which helps in verifying the user's condition after a fall event. The device operates on a **3.7V** rechargeable battery, making it portable, energy-efficient, and suitable for continuous wearable usage.

A user-friendly graphical interface designed using SquareLine Studio and implemented with the LVGL library provides real-time display of health data, system status, and alert notifications. When a fall is detected, the system prompts the user for confirmation through the display interface. If the user fails to respond within a predefined time, the system automatically sends an emergency alert to a caregiver via Wi-Fi communication, ensuring timely assistance.

The proposed system is compact, cost-effective, and privacy-friendly as it does not rely on camera-based monitoring. By integrating motion sensing, physiological monitoring, real-time processing, and IoT communication, the system offers a practical and efficient solution to improve the safety, independence, and quality of life of elderly individuals.

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ABBREVIATION

WHO	World Health Organisation	IDE	Integrated Development Environment
IoT	Internet of Things	ML	Machine Learning
GPS	Global Positioning System	CNN	Convolutional Neural Network
GSM	Global System for Mobile	SpO ₂	Peripheral Capillary Oxygen Saturation
ESP	Espress if Systems Platform	PPG	Photoplethysmography
SIM	Subscriber Identity Module	MPU	Microprocessor Unit
MAX30102	Maxim Integrated 30102		
SMS	Short Message Service		
WiFi	Wireless Fidelity		
LE	Low Energy		
SRAM	Static Random Access Memory		
UART	Universal Asynchronous Receiver Transmitter		
I2C	Inter Integrated Circuit		
SPI	Serial Peripheral Interface		
PWM	Pulse Width Modulation		
ADC	Analog to Digital Converter		
RGB	Red Green Blue		
LED	Light Emitting Diode		
USB	Universal Serial Bus		
IPS	In-Panel Switching		
Li Po	Lithium Polymer		
BAT	Battery		
SD	Secure Digital		
IC	Integrated Circuit		

INTRODUCTION

1.1 Problem overview

Falls are one of the most critical health concerns affecting elderly individuals and patients with mobility impairments. According to the World Health Organization (WHO), falls are the second leading cause of accidental or unintentional injury deaths worldwide, with a significant proportion occurring among people aged 65 years and above. The primary issue lies not only in the fall itself but in the delay in receiving timely assistance, which can result in serious complications such as fractures, head injuries, or even fatalities.

Existing fall detection systems, such as camera-based surveillance and commercial smartwatches, face several limitations. Camera-based systems compromise user privacy and are restricted to indoor environments, while smartwatches with built-in fall detection features are often expensive and inaccessible for most users. Moreover, many available systems lack portability and real-time alert mechanisms, making them unsuitable for continuous monitoring.

There is, therefore, a pressing need for a cost-effective, reliable, and privacy preserving solution capable of detecting falls in real time and notifying caregivers or family members immediately. The integration of Internet of Things (IoT) technology with wearable sensors offers a promising solution to address these issues. By combining accelerometer, gyroscope, and health monitoring sensors with wireless communication modules such as GSM and GPS, an intelligent IoT-based system can ensure immediate alerts, accurate location tracking, and enhanced safety for elderly individuals both at home and in healthcare facilities

1.2 Mini-project 2B objective

The primary objective of this mini-project is to design and develop an IoT-based smart wearable system capable of detecting falls in real time and providing immediate alerts to ensure rapid response. The system aims to improve the safety, independence, and overall quality of life of elderly individuals and patients who are at high risk of accidental falls due to age-related health issues or mobility limitations.

The project focuses on developing a reliable and compact wearable device using the ESP32-S3 microcontroller integrated with inertial sensors such as an accelerometer and gyroscope to continuously monitor motion and posture. To enhance the accuracy of fall detection, the system incorporates a MAX30102 sensor to monitor physiological parameters like heart rate and blood oxygen level (SpO₂), enabling verification of the user's condition after a fall event.

An IoT-based communication mechanism is implemented using wireless connectivity to transmit real-time alerts to caregivers or family members. The system is designed to be cost-effective, energy-efficient, and user-friendly, ensuring accessibility and long-term usability. Additionally, the device is developed in a wearable form factor with a lightweight design,

ensuring comfort, portability, and preservation of user privacy, as it does not rely on camera-based monitoring.

The project also includes testing and validation under different conditions to evaluate system performance, reduce false alarms, and ensure reliable detection of fall events. Furthermore, the design allows scope for future enhancements, such as integrating advanced algorithms or machine learning techniques for improved accuracy and predictive analysis.

1.3 Report outline

This report is organized into seven chapters to systematically present the development of the IoT-based wearable fall detection system.

Chapter 1 introduces the project by presenting the problem overview, objectives of the mini-project, and the overall structure of the report. It highlights the need for an efficient and reliable fall detection system for elderly individuals.

Chapter 2 provides a detailed literature review of existing fall detection systems, sensor technologies, and IoT-based healthcare solutions. It includes a review of related research papers and summarizes the limitations of current systems, which form the basis for the proposed work.

Chapter 3 explains the design and working principle of the system. It includes the block diagram, circuit diagram, and description of components along with their important specifications, giving a clear understanding of how the system is structured and operates.

Chapter 4 presents the simulation and results of the system. It covers the simulation of the accelerometer and gyroscope for fall detection, heart rate and SpO₂ monitoring, and discusses troubleshooting methods along with the solutions implemented.

Chapter 5 describes the software and GUI implementation of the project. It explains the use of Arduino IDE for programming, SquareLine Studio and LVGL for graphical user interface development, and 3D modeling using 3ds Max. It also covers the integration of software components and challenges faced during implementation.

Chapter 6 focuses on the hardware implementation of the system. It includes the hardware architecture, interfacing of components, power supply considerations, prototype development, system operation, and challenges encountered during hardware design.

Finally, Chapter 7 presents the conclusion and future work. It summarizes the outcomes of the project and discusses possible enhancements and improvements that can be made in the future.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The development of IoT-based fall detection systems has gained significant attention in recent years due to the increasing need for continuous monitoring and safety of elderly individuals. Falls are one of the leading causes of injury and mortality among senior citizens, especially when immediate medical assistance is not available. To address this issue, researchers have explored various technologies, including wearable sensors, wireless communication, and intelligent data processing techniques, to design efficient and reliable fall detection systems.

Recent studies focus on the use of low-cost wearable devices that integrate inertial sensors such as accelerometers and gyroscopes with biomedical sensors to enhance detection accuracy. IoT-enabled systems further improve the effectiveness of these devices by enabling real-time data transmission and instant alert generation to caregivers. Additionally, advancements in sensor fusion and machine learning techniques have contributed to reducing false alarms and improving system reliability in real-world scenarios.

Despite these developments, several challenges remain, including power constraints in wearable devices, maintaining user privacy, ensuring real-time response, and achieving accurate detection under different daily activities. This literature review aims to analyse existing research work, compare different approaches, and identify gaps in current systems, which form the basis for the proposed IoT-based smart fall detection watch.

2.2 Review of the papers

1. IoT-Based Smart Fall Detection Device for Elderly People – Gupta et al. (ICEECT 2024)

Gupta et al. [1] proposed an IoT-based smart fall detection device aimed at improving the safety and quality of life of elderly individuals. Their system utilized a wearable sensor based approach to continuously monitor body movements and detect sudden changes that indicate a fall. The device was built around a microcontroller unit integrated with an accelerometer and gyroscope, which together provided real-time motion data. Upon detecting abnormal acceleration beyond a preset threshold, the system automatically generated an alert through GSM and GPS modules, sending a text message containing the exact geographical location of the person to caregivers or family members. The project demonstrated how IoT technology can be effectively applied to ensure quick emergency response and remote monitoring. The authors emphasized that their model was cost effective, portable, and reliable, making it suitable for elderly care applications. Their work provided a strong foundation for low-cost, wearable fall detection systems and influenced subsequent designs that use multi-sensor integration and IoT communication for real-time alerts.

2. *Designing a Fall Detection System for Elderly People in Smart Healthcare IoT Using Multi-Sensor Data Fusion – Krishna et al. (ICAIQSA 2024)*

3.

Krishna et al. [2] introduced a multi-sensor data fusion approach to enhance the accuracy and reliability of fall detection systems within smart healthcare environments. Unlike traditional single-sensor methods, this design employed a fusion of accelerometer, gyroscope, and physiological sensors, allowing the system to more effectively differentiate between daily human activities and actual fall events. The authors implemented sensor fusion algorithms that combined motion and body parameter data to reduce false positives that commonly occur with simple threshold-based systems. Their approach ensured that minor movements such as sitting or bending were not misclassified as falls. The paper further discussed the use of IoT communication protocols to transmit alerts to healthcare servers and caregivers in real time. Experimental results showed a significant improvement in detection accuracy and system responsiveness, proving that multi-sensor data fusion provides a more dependable and practical solution for real-world fall detection. This paper influenced the design strategy of the current project, emphasizing the importance of combining motion and biomedical data for accurate and reliable fall detection.

4. *Recent Advances in Fall Detection and Prevention Using Machine Learning and IoT: A Comprehensive Review – Akilandeswari and Velusamy (ICETEA 2025)*

In a comprehensive review paper, Akilandeswari and Velusamy [3] analyzed the evolution of fall detection and prevention technologies, highlighting the integration of machine learning (ML) and IoT as the future of smart healthcare systems. The review covered a wide range of approaches from simple threshold-based algorithms to advanced machine learning and deep learning models such as Support Vector Machines (SVM), Random Forests, and Convolutional Neural Networks (CNN). The authors noted that while ML-based systems achieve higher accuracy, they require greater computational resources and energy consumption, which can be a limitation for battery-powered wearable devices. The study also discussed the trade-offs between camera-based systems, which provide high precision but raise privacy concerns, and wearable IoT-based systems, which are user friendly and protect personal privacy. The paper concluded that future research should focus on developing lightweight ML models optimized for embedded devices and improving energy efficiency in wearable IoT systems.

2.3 Summary of Literature Review

All three reviewed papers collectively establish a strong foundation for IoT-based fall detection systems. Gupta et al. demonstrated the feasibility of real-time IoT alert mechanisms using GSM and GPS, Krishna et al. improved detection accuracy through multi-sensor fusion, and Akilandeswari and Velusamy identified machine learning integration and energy efficiency as key future research directions. Together, these studies highlight the need for a system that is accurate, energy-efficient, cost-effective, and privacy preserving objectives that the current project aims to achieve by integrating motion sensing, biomedical monitoring, and IoT communication in a single compact wearable prototype.

CHAPTER 3

METHODOLOGY AND FUNCTIONING

3.1 Methodology

The circuit diagram of the proposed IoT-Based Smart System for Fall Detection which given in fig 3.1.1 is based on the continuous monitoring of a person's body motion and vital health parameters to accurately detect a fall and generate an automatic emergency alert. The system operates using a combination of motion sensors, biomedical sensors, and IoT communication modules integrated through the ESP32-S3 microcontroller, which serves as the central control unit.

The device uses Chronos-esp32 which is an open-source Arduino library and wrapper designed by fbiego that enables an ESP32 microcontroller to act as a smartwatch, interfacing with the Chronos App over BLE. It also allows us to build with ESP32 display for notification, time, weather and customize UI.

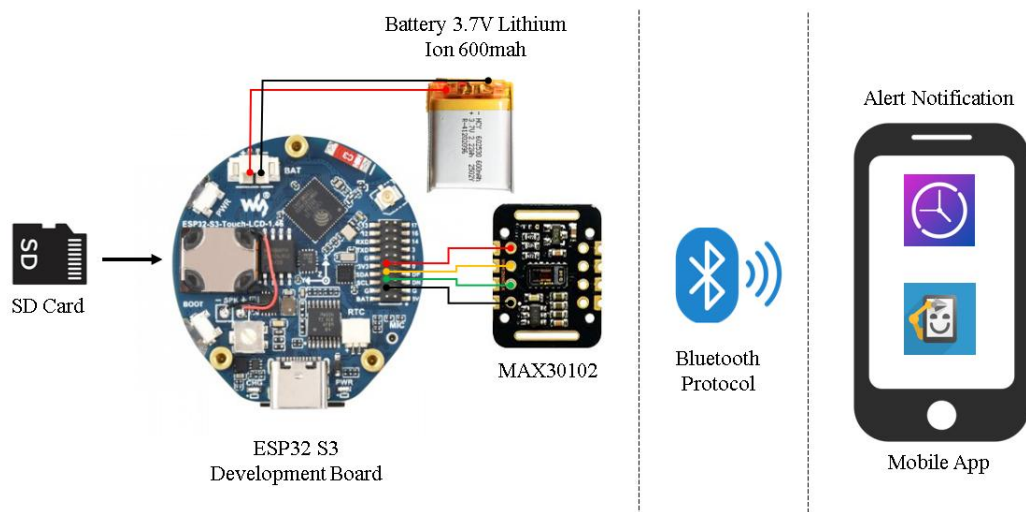


Fig 3.1.1: Circuit Diagram of Smart System for Fall Detection System

During normal operation, the accelerometer and gyroscope continuously sense linear acceleration and angular velocity corresponding to the user's daily activities such as walking, sitting, or standing. These readings are processed in real time by the microcontroller using a threshold-based algorithm that distinguishes normal motion from a sudden, rapid change in acceleration and orientation that typically occurs during a fall. When the acceleration magnitude or angular displacement exceeds the preset threshold, the system recognizes this pattern as a potential fall event.

When fall is detected then SOS protocol is initiated and print on display, after that the one button is assigned internally for to generating SOS signal & send to app. when fall is detected then Bluetooth disconnects and again reconnect which helps to send a trigger signal to the secondary app this app helps to send SMS to patients' caregiver with its GPS location.]

3.2 Components used, its important Specification.

1. ESP32-S31.46 Inch Round Display Development Board

In given Fig 3.2.1 is ESP32-S3 1.46-inch Round Display Development Board is a powerful microcontroller board based on the ESP32-S3 chip, designed for IoT, wearable, and smart display applications. It features a dual-core 32-bit Xtensa LX7 processor running up to 240 MHz, providing excellent performance for multitasking and sensor data processing. The board includes 512 KB of SRAM and 8–16 MB of Flash memory, allowing for efficient storage of programs and data. It supports both Wi-Fi (802.11 b/g/n) and Bluetooth 5.0 (LE) connectivity, enabling seamless wireless communication with mobile devices or cloud platforms. A key feature of this board is its 1.46-inch round IPS touchscreen display with a resolution of 172×320 pixels, which allows for clear and interactive visualization of data, system status, or user interfaces.

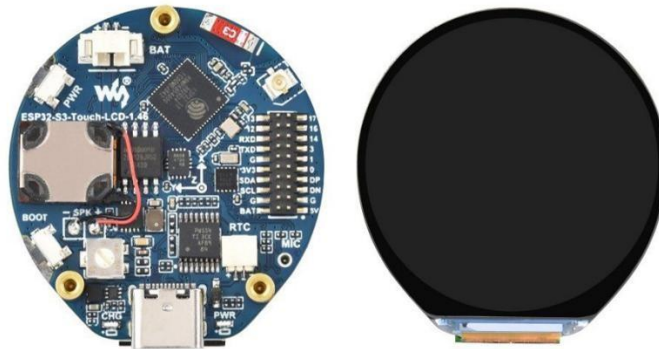


Fig 3.2.1: ESP32-S3 Development Board

The board operates on a 3.3V logic level and can be powered via a USB Type-C connector or a 3.7V lithium battery, making it ideal for portable and battery-powered projects. It includes an onboard battery charging circuit, allowing direct connection of a Li-ion or Li-Po battery through the BAT pin. Multiple communication interfaces such as UART, SPI, I²C, PWM, and ADC are available, making it highly compatible with various sensors and modules. Additionally, the board integrates a microSD card slot, RGB LED, power management IC, and reset/boot buttons for enhanced functionality and ease of use. The touch-enabled round display is particularly useful for user-interactive projects, such as wearable devices, health monitoring systems, and smart home controllers. Overall, the ESP32-S3 1.46-inch round display development board offers a compact, versatile, and powerful platform for developing IoT and embedded systems with an attractive graphical interface.

2. MAX30102 Hear Rate Sensor

The Fig 3.2.2 is MAX30102 sensor module which is integrated with pulse oximeter and heart-rate monitoring sensor designed for accurate detection of blood oxygen saturation (SpO₂) and pulse rate. It works on the principle of photoplethysmography (PPG), using two LEDs one

red (660 nm) and one infrared (880 nm) along with a photodetector to measure the amount of light absorbed by blood as it passes through a fingertip or earlobe.



Fig 3.2.2: MAX30102 Heart Rate & SpO₂ Sensor

The amount of light absorption varies with the oxygen level in the blood and the heart's pulsation, allowing the device to calculate both heart rate and SpO₂ values. The MAX30102 communicates with the microcontroller via the I²C interface, making it simple to integrate with boards like the ESP32-S3. It operates at a low voltage of 1.8V to 3.3V, ensuring minimal power consumption ideal for portable and wearable health monitoring systems. The sensor includes an internal temperature sensor for calibration, a low-noise analog signal processing circuit, and ambient light cancellation for accurate readings even in varying lighting conditions. In this fall detection system, the MAX30102 can be used to continuously monitor the elderly person's pulse and oxygen levels, helping assess their health status after a fall or during normal activity.

3.3 Functioning

The fig 3.3.1 shows development of the smartwatch firmware is carried out using Platform IO IDE integrated with Visual Studio Code, which provides a powerful and flexible environment for embedded system programming. The Arduino framework is used to simplify hardware interfacing and firmware development, while the core programming is implemented using C and C++ languages. Several important libraries are utilized to enhance functionality and reduce development complexity. The LVGL (Light and Versatile Graphics Library) is used to design an interactive and visually appealing graphical user interface. The ChronosESP32 library enables smartwatch-specific features and supports Bluetooth communication. The NimBLE library is used for efficient Bluetooth Low Energy communication, which is essential for low-power wearable devices. Additionally, the QMI8658 sensor library is used to interface with the accelerometer and gyroscope for motion sensing[6]. On the smartphone side, the Automate application is used to implement automation tasks such as detecting triggers, retrieving GPS location, generating Google Maps links, and sending SMS alerts automatically.

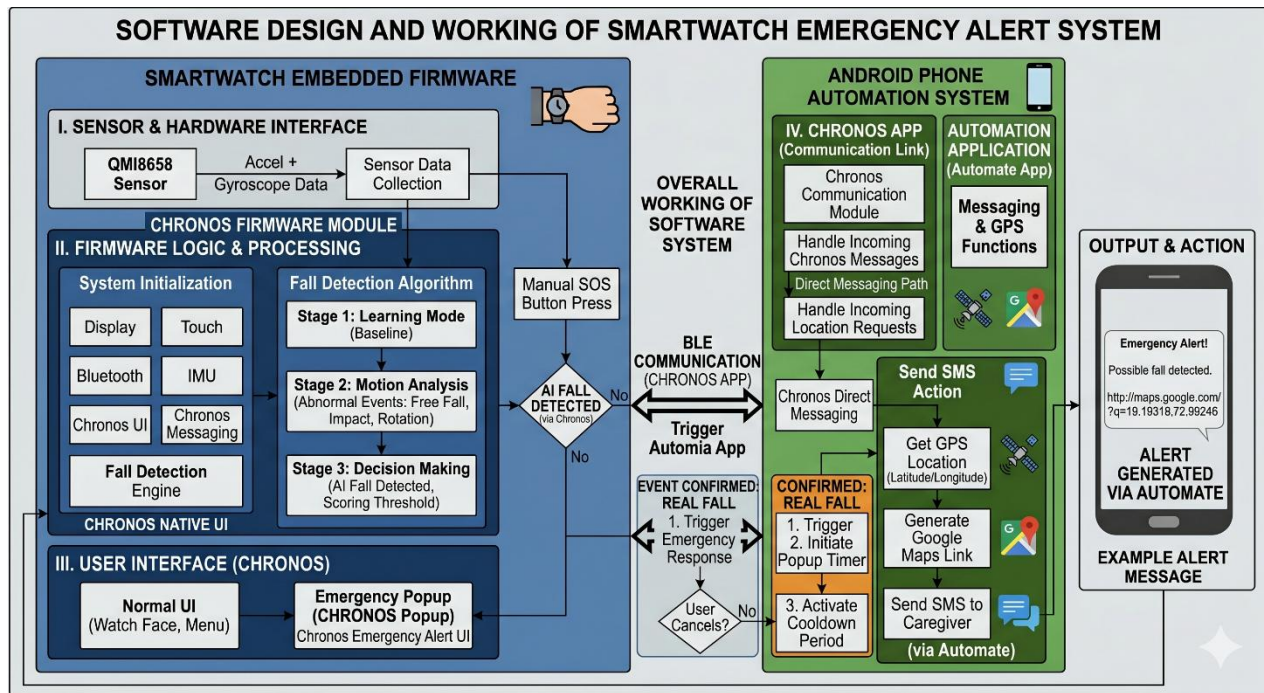


Fig 3.3.1 Workflow of Smart System for Fall Detection System

3.3.1 Overall Working of the Software System

As shown in fig 3.4.1 the software system operates based on an event-driven workflow. When an emergency event such as a fall is detected or when the user manually activates the SOS feature, the smartwatch displays an emergency popup with a countdown timer. During this period, the user has the option to cancel the alert if it is a false trigger. If the user does not respond within the given time, the system automatically proceeds to send an emergency signal to the connected smartphone via Bluetooth Low Energy. The smartphone then retrieves the current GPS location, generates a Google Maps link using the coordinates, and sends an SMS alert to the predefined caregiver. This structured workflow ensures both user control and automatic emergency handling.

3.3.2 Smartwatch Firmware Working

When the smartwatch is powered on, the firmware begins by initializing all hardware components including the display, touch interface, Bluetooth module, and IMU sensor. After successful initialization, the graphical user interface is loaded, and the fall detection algorithm is activated. The system then enters continuous monitoring mode. The QMI8658 sensor continuously provides real-time acceleration and angular velocity data along the X, Y, and Z axes. This data is processed by the firmware to analyze the user's motion patterns.

The fall detection mechanism used in the smartwatch is based on a multi-stage state machine that analyzes motion data from the IMU sensor. The system continuously samples acceleration and gyroscope data every 100 ms and calculates their magnitudes to monitor user movement. The detection process consists of three stages: free fall detection, impact detection, and stillness confirmation. In the first stage, a sudden drop in acceleration indicates a possible free fall. In the second stage, a sharp increase in acceleration confirms an impact. Finally, the system checks for stillness by ensuring stable acceleration and low angular movement, which

verifies the fall and reduces false alarms. Once a fall is confirmed, the smartwatch displays an emergency alert with a countdown timer. If the user does not cancel it, an emergency signal is sent to the smartphone, and a cooldown period is applied to avoid repeated alerts.

3.3.3 User Interface Working

The smartwatch user interface is developed using the LVGL graphics library, which allows smooth and responsive interaction. Under normal conditions, the interface displays features such as a digital watch face, menu navigation, and application screens. In case of an emergency, a dedicated popup screen is displayed showing a warning message along with a countdown timer. The user is given an option to cancel the alert within the countdown duration. If no action is taken, the system automatically proceeds with the emergency process. This design ensures both usability and safety by minimizing false alerts while maintaining responsiveness.

3.3.4 Manual SOS Function

In addition to automatic fall detection, the system provides a manual SOS feature that allows the user to request help at any time. When the user presses the SOS button on the smartwatch, an emergency popup is displayed, and a countdown timer is initiated. If the user does not cancel the request, the smartwatch sends an emergency trigger to the smartphone. This feature is particularly useful in situations where the user is conscious but requires immediate assistance.

3.3.5 Communication Between Watch and Phone

The communication between the smartwatch and the smartphone is established using Bluetooth Low Energy technology. BLE is chosen due to its low power consumption, fast connectivity, and suitability for wearable devices. It ensures reliable and continuous communication without significantly affecting battery life. The primary purpose of BLE communication in this system is to transmit emergency trigger signals from the smartwatch to the smartphone and maintain a stable connection for real-time operation.

3.3.6 Android Phone Automation Working

The Android smartphone uses the Automate application to execute predefined actions when an emergency trigger is received from the smartwatch. Upon receiving the signal, the application automatically retrieves the device's current GPS location, extracts the latitude and longitude coordinates, and generates a Google Maps link. This link is then included in an SMS message and sent to the caregiver. The entire process is executed without requiring any user intervention, ensuring quick and efficient emergency response.

3.3.7 GPS Location and Alert System

The smartphone's GPS module is used to obtain real-time location data, including latitude and longitude values. These coordinates are converted into a Google Maps URL format, which allows the caregiver to directly access the user's location through a clickable link. Once the link is generated, an SMS alert containing the emergency message and location link is sent to predefined contacts. This feature ensures that help can be provided quickly and accurately.

CHAPTER 4

SOFTWARE AND GUI IMPLEMENTATION

4.1 Introduction

The software implementation of the proposed IoT-based wearable smart watch plays a crucial role in enabling real-time data processing, user interaction, and communication. The system integrates embedded programming, graphical user interface (GUI) design, and 3D modeling to create a complete and user-friendly solution. Various software tools such as Arduino IDE, SquareLine Studio, and 3ds Max are used to develop, visualize, and implement different aspects of the system.

4.2 Square Line Studio for GUI Design (Phase 1)

The fig 4.2.1 is GUI design make in squareline studio. SquareLine Studio is used as the primary tool for designing the graphical user interface (GUI) of the wearable smart watch. It is specifically developed to work with the LVGL (Light and Versatile Graphics Library), which is widely used for creating embedded graphical interfaces on microcontrollers like the ESP32. The use of SquareLine Studio simplifies the process of designing complex and interactive user interfaces without writing extensive graphical code manually.

In this project, SquareLine Studio is utilized to create a visually appealing and user-friendly interface suitable for a smartwatch environment. Since the target users are elderly individuals, the GUI is designed with a focus on simplicity, readability, and ease of interaction. Large fonts, clear icons, and minimal screen clutter are used to ensure that the information is easily understandable.

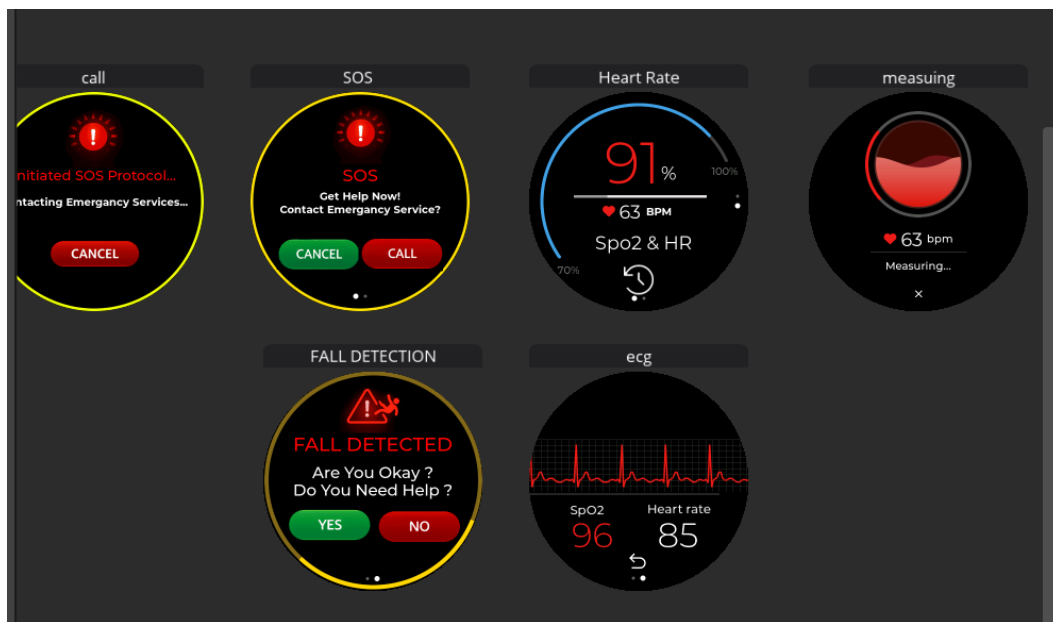


Fig 4.2.1 GUI Design on Square line Studio

The GUI design process begins by selecting the appropriate display resolution and screen layout that matches the 1.46-inch round display of the ESP32-S3 board. Various UI elements such as labels, buttons, icons, and containers are added using drag-and-drop features provided by SquareLine Studio. These elements are organized into different screens and layers to represent various system states.

Multiple screens are designed to represent different functionalities of the system, such as:

- Home Screen: Displays basic information like time and system status
- Health Monitoring Screen: Shows real-time heart rate and SpO₂ values
- Alert Screen: Displays emergency messages such as “Are you OK?” when a fall is detected
- Status Indicators: Show connectivity and system activity

Animations and transitions are also incorporated to enhance the user experience. For example, smooth transitions between screens and blinking alert messages are used to attract the user’s attention during emergency situations.

Once the GUI design is completed, SquareLine Studio generates the corresponding LVGL-compatible C code. This code includes all graphical elements, layouts, and configurations required to display the interface. The generated code is then exported and integrated into the Arduino IDE, where it is combined with the main program logic.

The advantage of using SquareLine Studio lies in its ability to reduce development time and simplify GUI design. Instead of manually coding each graphical component, developers can visually design the interface and directly generate optimized code. This also helps in maintaining consistency and improving the overall quality of the user interface.

4.3 3D Modelling using 3ds Max

3D modelling plays an important role in visualizing the physical design and structure of the proposed wearable smart watch. In this project, Autodesk 3ds Max is used to create a realistic and detailed model of the device, helping in understanding the placement of components and overall form factor before actual hardware fabrication.



Fig 4.3.1 3D Model of our Project

The fig 4.3.1 is our project 3D model modeling process begins with designing the outer casing of the smartwatch, considering factors such as size, shape, and ergonomics. Since the device is intended to be worn continuously, special attention is given to ensuring that the design is compact, lightweight, and comfortable for the user. A round casing is modeled to match the ESP32-S3 display module, with appropriate space allocated for internal components such as the battery, sensors, and circuit connections.

Different parts of the model are created separately and then assembled to form the complete structure. This includes the display housing, sensor slots, battery compartment, and strap attachment points. Proper dimensions are maintained to ensure that all components fit accurately within the enclosure. Ventilation and openings are also considered for sensor exposure, especially for the MAX30102 sensor, which requires direct contact or proximity to the user's skin for accurate readings.

Material properties, textures, and colors are applied to the model to provide a realistic appearance of the final product. This helps in presenting the design effectively during project demonstrations and reports. Additionally, the 3D model allows for easy modifications, enabling design improvements without physical rework.

The developed 3D design can further be used for 3D printing, allowing the creation of a prototype enclosure for real-world testing. This step bridges the gap between conceptual design and physical implementation.

Overall, the use of 3ds Max enhances the design process by providing a clear visualization of the wearable device, ensuring proper component placement, improving ergonomics, and supporting prototype development.

4.4 Software Architecture of the Proposed System (Phase 2)

We use phase 2 in our project because it is more reliable than before and much easier to connect it with our mobile phone.

4.4.1 Chronos App



Fig 4.4.1 Chronos App

The fig 4.4.1 shows the Chronos application is used as a companion interface for the ESP32-based smartwatch, enabling communication and control through Bluetooth Low Energy (BLE) [5]. It is specifically designed to work with ESP32 smartwatches and provides features similar to commercial wearable devices. In this project, the Chronos app helps establish a stable connection between the smartwatch and the smartphone, allowing data exchange and remote triggering of actions. It supports functionalities such as notifications, device synchronization, and communication handling. The app plays an important role in enabling the smartwatch to send signals (such as emergency triggers) to the smartphone efficiently. Its lightweight design and compatibility with BLE make it suitable for low-power wearable systems.

4.4.2 Automate App

The fig 4.4.2 is of Automate app. It is an Android-based automation tool used to perform predefined tasks automatically based on triggers. In this project, it is used to handle emergency alerts generated by the smartwatch. When the smartwatch sends a trigger signal via Bluetooth, the Automate app detects this event and executes a sequence of actions without user intervention. These actions include retrieving the current GPS location of the user, generating a Google Maps link using latitude and longitude coordinates, and sending an SMS alert to a predefined caregiver or emergency contact. The app uses a flowchart-based interface, making it easy to design and customize automation processes. Its ability to integrate multiple functions such as Bluetooth detection, GPS access, and messaging makes it a key component in ensuring a fast and reliable emergency response system.

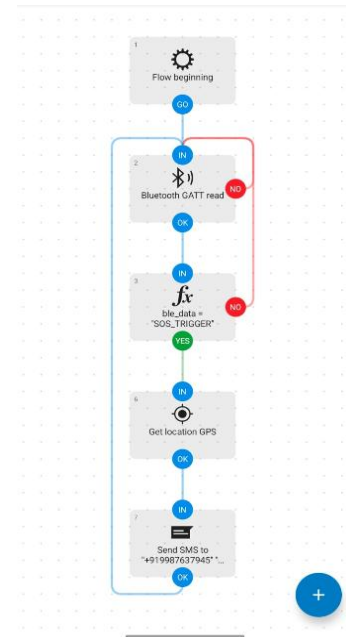


Fig 4.4.2 Automate App

4.5 Challenges and Solutions

During the development of the smartwatch emergency alert system, several technical challenges were encountered in both the hardware and software stages. These issues were systematically analyzed and resolved through debugging, testing, and optimization techniques.

One of the major problems faced was **inaccurate fall detection**. Initially, the system generated false triggers during normal hand movements and sometimes failed to detect actual falls. This issue was resolved by redesigning the detection algorithm into a multi-stage adaptive system that considers parameters such as free fall, impact, rotation, and stillness. Threshold values were fine-tuned through repeated real-time testing to improve accuracy.

Another challenge was related to **BLE (Bluetooth Low Energy) communication reliability** between the smartwatch and the smartphone. Attempts to trigger alerts using direct BLE commands and device name changes were inconsistent due to limitations of the ESP32-S3 and conflicts with the Chronos framework. This problem was addressed by shifting to a more reliable approach using phone-side automation, where a specific event (Bluetooth connection trigger) is detected by the mobile application to initiate emergency actions.

The project also encountered **memory management issues**, particularly errors such as “malloc failed” and buffer allocation failures. These were caused by excessive use of applications and background processes in the firmware. To solve this, unnecessary application modules (such as weather, navigation, and demo apps) were removed, reducing RAM usage and stabilizing the system.

Another issue was the **unintended triggering of SMS alerts during normal Bluetooth connection events**. This resulted in false emergency messages being sent. The problem was mitigated by redesigning the trigger logic so that alerts are only sent after the SOS confirmation timeout or manual activation, ensuring that only genuine emergencies generate alerts.

There were also challenges in **UI integration and responsiveness**. Initially, the SOS popup did not display correctly or failed to override other screens. This was resolved by restructuring the UI control logic and ensuring proper function calls to force the popup screen to the foreground using the LVGL-based system.

Additionally, **GPS acquisition delays** on the smartphone side caused slower alert generation. This was improved by adjusting the location provider settings (Balanced/GPS mode) in the automation app, ensuring faster and more reliable location retrieval.

Overall, these challenges were addressed through iterative testing, debugging using the serial monitor, modular code improvements, and optimization of both firmware and mobile automation logic. The troubleshooting process significantly enhanced the system's reliability, accuracy, and real-world usability.

4.6 Summary

The software and GUI implementation of the proposed IoT-based wearable smart watch successfully integrates multiple technologies to create a functional and efficient system. The Arduino IDE enables the development of embedded programs for sensor interfacing, data processing, and communication. SquareLine Studio simplifies the design of an intuitive and user-friendly graphical interface, while the LVGL library ensures smooth and efficient GUI rendering on the ESP32 platform.

The integration of these software components allows real-time interaction between hardware and user interface, enabling continuous monitoring of motion and physiological parameters. The system is capable of detecting fall events, verifying user condition, and generating alerts with minimal delay. The GUI enhances user experience by providing clear and interactive feedback, making the device easy to use, especially for elderly individuals.

Despite challenges such as sensor noise, memory limitations, and communication issues, effective solutions were implemented to ensure reliable system performance. The final implementation demonstrates a balanced combination of functionality, efficiency, and usability.

Overall, the software and GUI implementation plays a critical role in achieving the objectives of the project, providing a robust and practical solution for real-time fall detection and elderly care monitoring.

CHAPTER 5

RESULT & ANALYSIS

RESULTS

The fig 5.1 show the smartwatch-based emergency alert system. Which was successfully developed and tested, demonstrating reliable performance in real-time conditions. The graphical user interface designed using SquareLine Studio was successfully implemented on the hardware using the LVGL framework. All UI components, including the main display and SOS popup screen, functioned correctly with smooth touch interaction and proper visual rendering. The SOS popup was triggered accurately during both automatic fall detection and manual activation, and the countdown timer operated as expected.



Fig 5.1 Emergency SOS popup screen



Fig 5.2 Countdown Timer after Fall Detection

The fig 5.2 shows fall detection system. That was tested under various motion conditions such as normal movement, sudden impact, and simulated falls. The system successfully detected actual fall events while ignoring normal activities, indicating improved detection accuracy. In fig 5.3 the addition to fall detection system we also monitored the user's heart rate using the onboard sensor. The heart rate data was successfully acquired and displayed, demonstrating the system's capability to monitor basic physiological parameters.

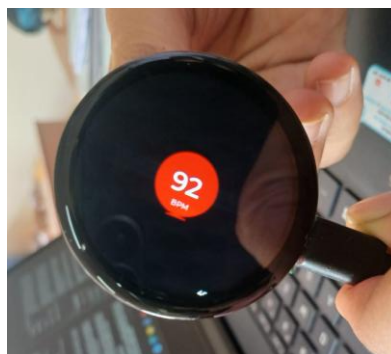


Fig 5.3 Heart Rate Monitoring

The emergency alert mechanism functioned correctly by initiating an alert after the countdown period if the user did not cancel it. The Bluetooth connection between the smartwatch and smartphone was successfully established, enabling communication between devices. In the fig 5.4 we can see that the system retrieved real-time GPS coordinates from the smartphone, generated a Google Maps link, and sent it via SMS to the caregiver. The received message correctly displayed the user's location on Google Maps. Overall, the system performed all intended operations, including fall detection, heart rate monitoring, SOS alert generation, location tracking, and emergency communication.

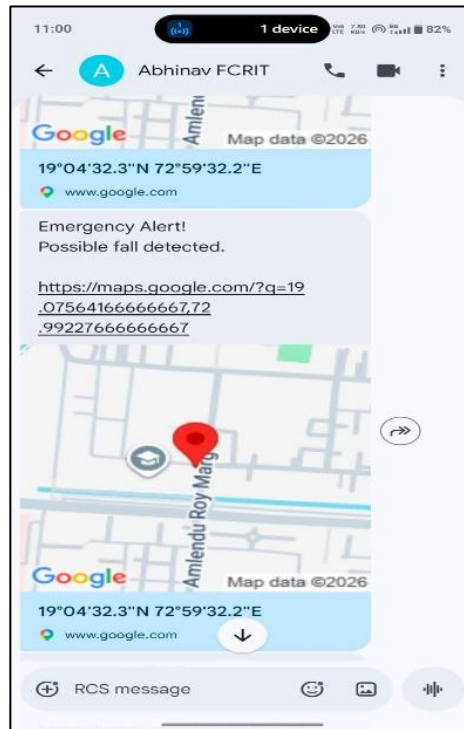


Fig 5.4 GPS Coordinates sent via SMS

ANALYSIS

The analysis of the system performance indicates that the integration of motion sensing, physiological monitoring, and smartphone automation provides a comprehensive emergency response solution. The adaptive fall detection algorithm improved accuracy by considering multiple parameters such as acceleration, impact, rotation, and stillness, thereby reducing false alarms compared to simple threshold-based approaches.



Fig 5.5 Final Prototype of Smart System for Fall Detection

The inclusion of heart rate monitoring adds an important health dimension to the system, allowing continuous observation of the user's physical condition. This feature enhances the system's usefulness, especially in scenarios where abnormal heart activity may accompany or precede a fall event.

The GUI simulation using SquareLine Studio helped validate the design before implementation, resulting in a responsive and user-friendly interface. The SOS popup mechanism effectively prevented false alerts by allowing user intervention before triggering emergency communication.

Although Bluetooth Low Energy communication was successfully established, direct BLE-based triggering showed limitations due to system constraints and memory usage. This was overcome by implementing smartphone-based automation, which provided a more stable and reliable triggering mechanism. Additionally, memory optimization by removing unnecessary applications improved system performance and reduced runtime errors.

The GPS and SMS functionalities were reliable, providing accurate location data and timely communication to the caregiver. Minor delays in GPS acquisition were observed initially but were minimized through optimization of location settings. Overall, the system achieved a balance between accuracy, responsiveness, and reliability.

The final system demonstrates effective integration of fall detection, heart rate monitoring, and emergency communication, making it a practical and efficient solution for real-world personal safety applications.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

The mini project titled “IoT-Based Smart System for Fall Detection” aims to design and develop a smart, wearable device capable of detecting accidental falls and providing immediate alerts to caregivers or family members. The system is specifically designed to enhance the safety and independence of elderly individuals and patients with mobility challenges, for whom falls often pose serious health risks. The project integrates multiple sensors and IoT communication modules to create an affordable, reliable, and portable solution for real-time health monitoring.

The prototype is built around the ESP32-S3 microcontroller, which serves as the central processing unit. It interfaces with an accelerometer and gyroscope to track body motion, a MAX30102 sensor to monitor heart rate and oxygen saturation levels, and GSM (SIM800L) and GPS (NEO-6M) modules to send alert messages containing real-time location data. During normal operation, the system continuously measures and displays the user’s physiological parameters. When a sudden fall or abnormal movement is detected based on the sensor threshold values the system immediately activates a buzzer alarm and transmits an SMS alert to the registered contact with the user’s location coordinates.

Simulation results validated the functionality and accuracy of the proposed design. The system effectively differentiated between normal activities and fall events, generating rapid and reliable alerts without false triggers. Its cost-effectiveness, low power consumption, and privacy-preserving nature make it a strong alternative to expensive commercial smartwatches or camera-based monitoring systems.

In summary, this project successfully demonstrates how IoT and sensor fusion technologies can be applied to create an intelligent, real-time fall detection system that ensures quick response, user safety, and improved quality of life for elderly individuals. The developed model provides a foundation for future enhancements such as machine learningbased motion classification and mobile app integration for advanced healthcare monitoring.

6.2 Future work

Although the proposed IoT-based wearable fall detection system demonstrates reliable performance, there is significant scope for further enhancement and expansion to improve its functionality, accuracy, and usability.

One of the major improvements can be the integration of GPS and GSM modules, which will enable real-time location tracking and ensure that emergency alerts are sent even in the absence of Wi-Fi connectivity. This will make the system more effective for outdoor use and enhance its reliability in critical situations.

Another important area of enhancement is the implementation of machine learning algorithms for fall detection. Instead of relying only on threshold-based methods, advanced techniques such as classification models can be used to analyze motion patterns more accurately. This will help in reducing false alarms and improving the system's ability to distinguish between different types of activities and actual fall events.

The system can also be extended by developing a dedicated mobile application that allows caregivers to monitor the user's health parameters in real time. Features such as alert history, health data logging, and emergency contact management can be included to provide a more comprehensive monitoring solution.

Further improvements can be made in terms of power management and battery optimization. Techniques such as sleep modes, efficient sensor usage, and low-power communication protocols can be implemented to extend battery life, making the device more suitable for long-term continuous usage.

From a hardware perspective, miniaturization and improved wearable design can be achieved by optimizing the placement of components and using compact modules. This will enhance user comfort and make the device more practical for daily use.

Additionally, integration with cloud platforms can enable remote data storage, analysis, and access for healthcare professionals. This can support advanced features such as long-term health monitoring and predictive analytics.

Finally, future work may also include predictive fall detection, where the system identifies risky movements or patterns before an actual fall occurs, allowing preventive measures to be taken. This would significantly enhance the overall effectiveness of the system in ensuring user safety.

LEARNING OUTCOMES

ECMNP603-01 Identify problems based on societal or research needs and methodology for solving them.

ECMNP603-02 Apply knowledge and skills to solve societal problems collaboratively.

ECMNP603-03 Develop interpersonal skills necessary for teamwork.

ECMNP603-04 Analyse, verify, and validate results effectively through various methodologies, including, test cases/benchmark data/theoretical/inferences/experiments/simulations, etc.

ECMNP603-05 Evaluate the societal and environmental impacts of proposed solutions.

ECMNP603-06 Adhere to standard engineering practices

ECMNP603-07 Excel in written and oral communication by technical report writing, oral presentation, and publishing results.

ECMNP603-08 Gain technical competencies by participating in competitions, hackathons, etc.

ECMNP603-09 Demonstrate lifelong learning capabilities through self-directed group projects.

ECMNP603-10 Apply project management principles effectively.

Table 5.1: CO-PO Correlation

CO ID	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
ECMNP603-01	2	3				3					
ECMNP603-02			3			2					
ECMNP603-03								3	3		
ECMNP603-04		3	3	3							
ECMNP603-05						1					
ECMNP603-06							2				
ECMNP603-07									3		
ECMNP603-08	2				2				2	3	
ECMNP603-09											3
ECMNP603-10										3	

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<https://doi.org/10.1109/iceect61758.2024.10739038>
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<https://doi.org/10.1109/ICAIQSA64000.2024.10882340>
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- [4] <https://www.analog.com/media/en/technical-documentation/data-sheets/max30102.pdf>
- [5] <https://github.com/fbiego/chronos-esp32>
- [6] <https://github.com/fbiego/esp32-c3-mini>

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Parag Dhumal (3023170)
Abhinav Thokale (3023171)
Manisha Kunder (3022130)

APPENDIX

DATA SHEETS

1. MAX30102 [6]

Click [here](#) for production status of specific part numbers.

MAX30102

High-Sensitivity Pulse Oximeter and Heart-Rate Sensor for Wearable Health

General Description

The MAX30102 is an integrated pulse oximetry and heart-rate monitor module. It includes internal LEDs, photodetectors, optical elements, and low-noise electronics with ambient light rejection. The MAX30102 provides a complete system solution to ease the design-in process for mobile and wearable devices.

The MAX30102 operates on a single 1.8V power supply and a separate 3.3V power supply for the internal LEDs. Communication is through a standard I²C-compatible interface. The module can be shut down through software with zero standby current, allowing the power rails to remain powered at all times.

Applications

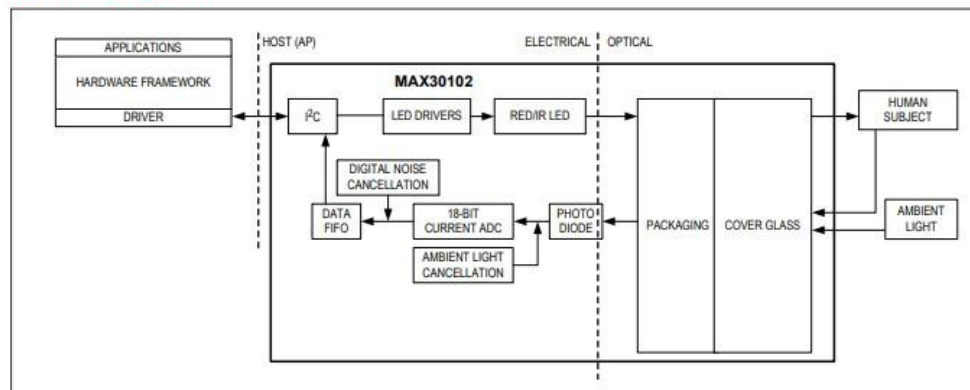
- Wearable Devices
- Fitness Assistant Devices
- Smartphones
- Tablets

Benefits and Features

- Heart-Rate Monitor and Pulse Oximeter Sensor in LED Reflective Solution
- Tiny 5.6mm x 3.3mm x 1.55mm 14-Pin Optical Module
 - Integrated Cover Glass for Optimal, Robust Performance
- Ultra-Low Power Operation for Mobile Devices
 - Programmable Sample Rate and LED Current for Power Savings
 - Low-Power Heart-Rate Monitor (< 1mW)
 - Ultra-Low Shutdown Current (0.7μA, typ)
- Fast Data Output Capability
 - High Sample Rates
- Robust Motion Artifact Resilience
 - High SNR
- -40°C to +85°C Operating Temperature Range

Ordering Information appears at end of data sheet.

System Diagram



19-7740; Rev 1; 10/18



Power Sequencing and Requirements

Power-Up Sequencing

Figure 6 shows the recommended power-up sequence for the MAX30102.

It is recommended to power the V_{DD} supply first, before the LED power supplies (V_{LED+}). The interrupt and I²C pins can be pulled up to an external voltage even when the power supplies are not powered up.

After the power is established, an interrupt occurs to alert the system that the MAX30102 is ready for operation. Reading the I²C interrupt register clears the interrupt, as shown in Figure 6.

Power-Down Sequencing

The MAX30102 is designed to be tolerant of any power supply sequencing on power-down.

I²C Interface

The MAX30102 features an I²C/SMBus-compatible, 2-wire serial interface consisting of a serial data line (SDA) and a serial clock line (SCL). SDA and SCL facilitate communication between the MAX30102 and the master at clock rates up to 400kHz. Figure 1 shows the 2-wire interface timing diagram. The master generates SCL and initiates data transfer on the bus. The master device writes data to the MAX30102 by transmitting the proper slave address followed by data. Each transmit sequence is framed by a START (S) or REPEATED START (Sr) condition and a STOP (P) condition. Each word transmitted to the MAX30102 is 8 bits long and is followed by an acknowledge clock pulse. A master reading data from the MAX30102 transmits the proper slave address followed by a series of nine SCL pulses.

The MAX30102 transmits data on SDA in sync with the master-generated SCL pulses. The master acknowledges receipt of each byte of data. Each read sequence is framed by a START (S) or REPEATED START (Sr) condition, a not acknowledge, and a STOP (P) condition. SDA operates as both an input and an open-drain output. A pullup resistor, typically greater than 500Ω, is required on SDA. SCL operates only as an input. A pullup resistor, typically greater than 500Ω, is required on SCL if there are multiple masters on the bus, or if the single master has an open-drain SCL output. Series resistors in line with SDA and SCL are optional. Series resistors protect the digital inputs of the MAX30102 from high voltage spikes on the bus lines and minimize crosstalk and undershoot of the bus signals.

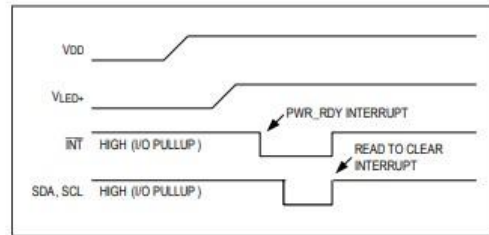


Figure 6. Power-Up Sequence of the Power Supply Rails

Bit Transfer

One data bit is transferred during each SCL cycle. The data on SDA must remain stable during the high period of the SCL pulse. Changes in SDA while SCL is high are control signals. See the [START and STOP Conditions](#) section.

START and STOP Conditions

SDA and SCL idle high when the bus is not in use. A master initiates communication by issuing a START condition. A START condition is a high-to-low transition on SDA with SCL high. A STOP condition is a low-to-high transition on SDA while SCL is high (Figure 7). A START condition from the master signals the beginning of a transmission to the device. The master terminates transmission, and frees the bus, by issuing a STOP condition. The bus remains active if a REPEATED START condition is generated instead of a STOP condition.

Early STOP Conditions

The MAX30102 recognizes a STOP condition at any point during data transmission except if the STOP condition occurs in the same high pulse as a START condition. For proper operation, do not send a STOP condition during the same SCL high pulse as the START condition.

Slave Address

A bus master initiates communication with a slave device by issuing a START condition followed by the 7-bit slave ID. When idle, the MAX30102 waits for a START condition followed by its slave ID. The serial interface compares each slave ID bit by bit, allowing the interface to power down and disconnect from SCL immediately if an incorrect slave ID is detected. After recognizing a START condition followed by the correct slave ID, the MAX30102 is programmed to accept or send data. The LSB of the slave ID word is the read/write (R/W) bit. R/W indicates whether the master is writing to or reading data from the MAX30102 ($R/\bar{W} = 0$ selects a write condition, $R/\bar{W} = 1$ selects a read condition).

After receiving the proper slave ID, the MAX30102 issues an ACK by pulling SDA low for one clock cycle.

The MAX30102 slave ID consists of seven fixed bits, B7–B1 (set to 0b1010111). The most significant slave ID bit (B7) is transmitted first, followed by the remaining bits. Table 17 shows the possible slave IDs of the device.

Acknowledge

The acknowledge bit (ACK) is a clocked 9th bit that the MAX30102 uses to handshake receipt each byte of data when in write mode (Figure 8). The MAX30102 pulls down SDA during the entire master-generated 9th clock pulse if the previous byte is successfully received. Monitoring ACK allows for detection of unsuccessful data transfers. An unsuccessful data transfer occurs if a receiving device is busy or if a system fault has occurred. In the event of an unsuccessful data transfer, the bus master retries communication. The master pulls down SDA

during the 9th clock cycle to acknowledge receipt of data when the MAX30102 is in read mode. An acknowledge is sent by the master after each read byte to allow data transfer to continue. A not-acknowledge is sent when the master reads the final byte of data from the MAX30102, followed by a STOP condition.

Write Data Format

For the write operation, send the slave ID as the first byte followed by the register address byte and then one or more data bytes. The register address pointer increments automatically after each byte of data received, so for example the entire register bank can be written by at one time. Terminate the data transfer with a STOP condition. The write operation is shown in Figure 9.

The internal register address pointer increments automatically, so writing additional data bytes fill the data registers in order.

Table 17. Slave ID Description

B7	B6	B5	B4	B3	B2	B1	B0	WRITE ADDRESS	READ ADDRESS
1	0	1	0	1	1	1	R/W	0xAE	0xAF

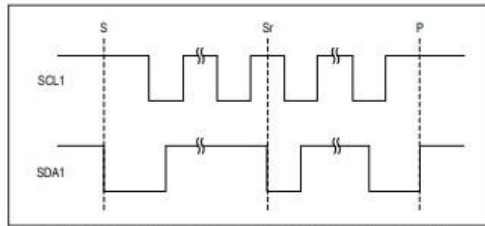


Figure 7. START, STOP, and REPEATED START Conditions

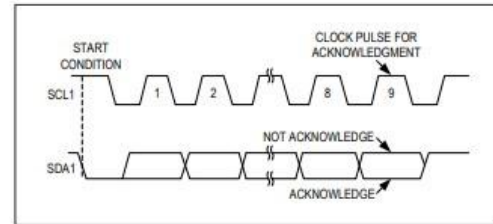


Figure 8. Acknowledge

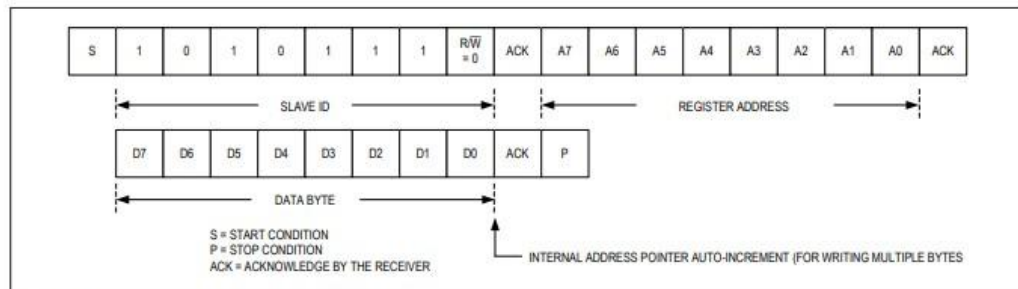
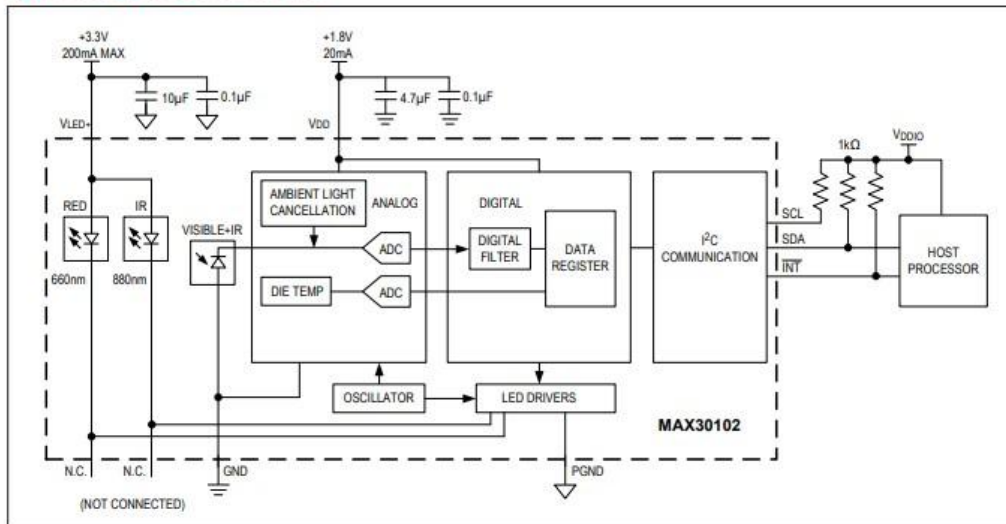


Figure 9. Writing One Data Byte to the MAX30102

Typical Application Circuit



Ordering Information

PART	TEMP RANGE	PIN-PACKAGE
MAX30102EFD+T	-40°C to +85°C	14-Lead OESIP (0.8mm Pin Pitch)

+Denotes lead(Pb)-free/RoHS-compliant package.

T = Tape and reel.

